

Wheeled inverted pendulum type assistant robot: design concept and mobile control

Seonghee Jeong · Takayuki Takahashi

Received: 10 December 2007 / Accepted: 7 May 2008 / Published online: 29 May 2008
© Springer-Verlag 2008

Abstract This paper describes the design concept of the human assistant robot I-PENTAR (Inverted PENDulum Type Assistant Robot) aiming at the coexistence of safety and work capability and its mobile control strategy. I-PENTAR is a humanoid type robot which consists of a body with a waist joint, arms designed for safety, and a wheeled inverted pendulum mobile platform. Although the arms are designed low-power and lightweight for safety, it is able to perform tasks that require high power by utilizing its self-weight, which is the feature of a wheeled inverted pendulum mobile platform. I-PENTAR is modeled as a three dimensional robot; with controls of inclination angle, horizontal position, and steering angle to achieve high mobile capability. The motion equation is derived considering the non-holonomic constraint of the two-wheeled mobile robot, and a state feedback control method is applied for basic mobile controls wherein the control gain is calculated by the LQR method. Through several experiments of balancing, linear running, and steering, it was confirmed that the robot could realize stable mobile motion in a real environment by the proposed controller.

Keywords Inverted pendulum robot · Mobile control · Assist robot

1 Introduction

1.1 Background

Recently, many robots have been developed for providing physical assistance to humans in various activities such as delivery and cleaning. It is essential that robots with high work capability provide not only various services but also high safety without causing any injury. However, since the two functions possess characters that mutually conflict with each other, it is difficult to reconcile them. For example, although safety is usually realized by making a lightweight, low speed, and low-power robot, this will cause a drop in its working capabilities such as loading and movement.

A robot with an unstable mobile platform, such as a bipedal or an inverted pendulum robot, can resolve this problem effectively by using its instability skillfully. Since these robots employ central of gravity (CoG) for the stable control of their system, they can perform work that demands high power with a low-power and lightweight arm by skillfully controlling the posture of the body mounted on it.

Among these robots, in this paper, we focus on the inverted pendulum type robot, which employs two wheels for its mobility. In addition to the above features, a two-wheeled inverted pendulum robot has some useful characteristics as a human-assistant robot. First, since it has a simpler structure, it contributes to reducing the weight of the system compared to the bipedal robot. Second, it ensures high safety against overturn when unpredictable collision occurs, since it is always controlled to maintain dynamic stability. Finally, since it has a mechanically small footprint and does not have any caster, it can achieve high mobile capability such as a quick response and high step traveling ability. From these features, a two-wheeled inverted pendulum type robot can

S. Jeong (✉) · T. Takahashi
Symbiotic Systems Science of Fukushima University,
Fukushima City, Japan
e-mail: jeong@sss.fukushima-u.ac.jp

T. Takahashi
e-mail: taka@sss.fukushima-u.ac.jp

be considered greatly suitable for supporting human work that requires high safety and work capability.

1.2 Related research

An inverted pendulum is mechanically unstable and it overturns when stabilization control is not applied. Thus, most studies on these robots aim at improving their stability; many such studies are still in progress. Recently, studies that consider these robots not as objects of stable control but as mobile robots have increased, for example, wheeled inverted pendulum robots.

Many studies on the wheeled inverted pendulum type robot focus on improving its mobile performance, such as running, steering, and trajectory following. Yamafuji [1–3] proposed a wheeled inverted pendulum type robot and realized stable traveling with it; Matumoto [4,5] exploited stable traveling of the robot by estimating its attitude using only internal sensors on a flat road as well as on an inclined path. Ha and Yuta [7,8] developed an inverted pendulum mobile robot that could traverse an indoor environment with high mobile performance and robustness. Although the robots in the above studies were modeled as a two-dimensional robot, certain studies have modeled them as a three-dimensional one in order to perform balancing, linear running, and steering control simultaneously [9,10,12–14].

On the other hand, a few studies have been conducted to realize not only movement but also task by the robot. Matumoto [6] realized a cooperative transportation task between two wheeled inverted pendulum robots by estimating external force and acting force by an observer. Bryan [15] recently proposed the pushing task that uses the robot's dynamic stability. Most recently, humanoid type robots with an inverted pendulum mobile has been proposed for providing powerful mobility and limited manipulation tasks [16,17] or for human friendly gestures [18].

1.3 Research target

As we can see from the above, most studies on the wheeled inverted pendulum robot focus on its balancing stability and mobile performance, and few studies consider its work capability as a human-assistant robot. The aim of this study is to develop an assistant robot with high safety and work capability employing the wheeled inverted pendulum mechanism, and to realize various motions, such as standing and sitting, picking up an object from ground, and lifting a heavy object. In this paper, we propose the prototype human-assistant robot, I-PENTAR (Inverted PENDulum Type Assistant Robot), and describe its design concept and mobile control in detail as the first step toward the goal.

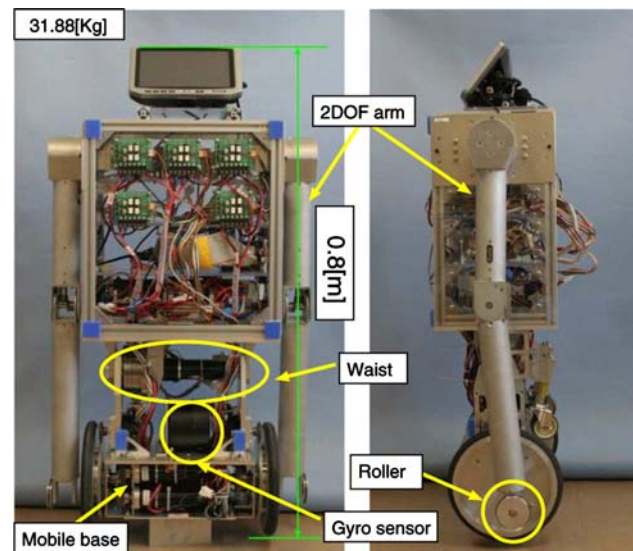


Fig. 1 Appearance of the inverted pendulum type assistant robot I-PENTAR

Table 1 General specifications of I-PENTAR

Size (m)	900 (H) × 340 (W) × 210 (D)
Weight (kg)	32
DOF	Mobile (2), Arm (4), Waist (1)
CPU	Pentium 3 500 (Mhz)
Sensor	Gyro sensor (1), Encoder (9)
Motor	DC motor (9)
Interface (inner)	FPGA(PC104), VGA monitor-1
OS	ARTLinux (sampling time: 1 (ms))
Devel. Language	C++

2 Inverted pendulum type assistant robot

Figure 1 shows the appearance of the prototype robot proposed in this study, I-PENTAR, and Table 1 describes the general specifications of the robot. To realize various services, I-PENTAR is composed of a body with a waist joint, two low powered and light arms, and a two-wheeled inverted pendulum type mobile platform.

It adopts a double motor method on its mobile actuator, which actuates a wheel with two motors in order to reduce the effect of backlash in balancing control and to increase power of the mobile platform. The lower body shares its axis with the wheel and freely rotates with respect to the axis, thereby realizing an inverted pendulum mechanism. The body is divided into the lower part and the upper part by a waist joint, which adopts a harmonic gear to reduce any backlash, and a rate gyro sensor is located in the lower body to detect an inclination angle with respect to the ground. There is a mechanical stopper at the elbow to execute a task that requires high power even if it does not have a high-power

Table 2 Parameters

θ_r, θ_l (rad)	Rotational angle of the right and left wheel
$\mathbf{P}_c$ (m)	Position of the vehicle in Σ_v
x_v (m)	Linear position of the vehicle
v (m/s)	Linear velocity of the vehicle
v_r, v_l (m/s)	Linear velocity of the left and right wheels
ψ (rad)	Inclination angle of CoG
ϕ (rad)	Steering angle of the vehicle
M_g (kg)	Mass of CoG
m_w (kg)	Mass of the wheel
l_g (m)	Length between the origin of vehicle fixed coordinates and CoG
b (m)	Distance between the origin of vehicle fixed coordinates and the center of wheel axis
r_w (m)	Radius of the wheel
I_g (kgm ²)	Moment of inertia of the body
I_{wa} (kgm ²)	Inertia of the wheel about wheel axis
I_{wd} (kgm ²)	Inertia of the wheel about its diameter
I_{ra} (kgm ²)	Inertia of the motor rotor about its axis
I_{rd} (kgm ²)	Inertia of the motor rotor about its diameter
γ	Reduction ratio
τ_r, τ_l (Nm)	Motor torque of the right and left wheels
c_r, c_l (Nm/(rad/s))	Viscosity coefficient of right and left wheel axle
$\mathbf{P}_g$ (m)	Position of CoG in Σ_o
$\mathbf{P}_v$ (m)	Position of the vehicle in Σ_o
$\mathbf{V}_g$ (m/s)	Velocity of CoG in Σ_o
$\mathbf{\Omega}_g$ (rad/s)	Angular velocity of CoG

3.2 3DOF motion equations

In this section, the motion equations of I-PENTAR for mobile control are derived. The position of CoG of the body, ${}^w\mathbf{P}_g$, and velocity, ${}^w\mathbf{V}_g$, with respect to the global coordinates is given as

$${}^w\mathbf{P}_g = \begin{bmatrix} x_g \\ y_g \\ z_g \end{bmatrix} = \begin{bmatrix} x_c + l_g \sin \psi \cos \phi \\ y_c + l_g \sin \psi \sin \phi \\ r_w + l_g \cos \psi \end{bmatrix} \quad (1)$$

$${}^w\mathbf{V}_g = \begin{bmatrix} \dot{x}_c + l_g(\dot{\psi} \cos \psi \cos \phi - \dot{\phi} \sin \psi \sin \phi) \\ \dot{y}_c + l_g(\dot{\psi} \cos \psi \sin \phi + \dot{\phi} \sin \psi \cos \phi) \\ -l_g \dot{\psi} \sin \psi \end{bmatrix} \quad (2)$$

and the angular velocities of the mobile base and CoG are given as

$$\mathbf{\Omega}_v = [0 \ 0 \ \dot{\phi}] \quad (3)$$

$$\mathbf{\Omega}_g = [-\dot{\phi} \sin \psi \ \dot{\psi} \ \dot{\phi} \cos \psi] \quad (4)$$

The kinetic energy of the wheels and body, T_g and T_w , respectively, damping energy, D , potential energy, U_g , can be

derived using Eqs. (1)–(4) as follows;

$$\begin{aligned} T_g &= \frac{1}{2} M_g \mathbf{V}_g^T \mathbf{V}_g + \frac{1}{2} \mathbf{\Omega}_g^T \mathbf{I}_g \mathbf{\Omega}_g \\ &= \frac{M_g}{2} [\dot{x}_c^2 + 2\dot{x}_c l_g (\dot{\psi} \cos \psi \cos \phi - \dot{\phi} \sin \psi \sin \phi) \\ &\quad + \dot{y}_c^2 + 2\dot{y}_c l_g (\dot{\psi} \cos \psi \sin \phi + \dot{\phi} \sin \psi \cos \phi) \\ &\quad + l_g^2 \dot{\psi}^2 + l_g^2 \dot{\phi}^2 \sin^2 \psi] + \frac{1}{2} (I_{xx} \dot{\phi}^2 \sin^2 \psi \\ &\quad + I_{yy} \dot{\psi}^2 + I_{zz} \dot{\phi}^2 \cos^2 \psi) \end{aligned} \quad (5)$$

$$\begin{aligned} T_w &= \frac{1}{2} m_w r_w^2 (\dot{\theta}_r^2 + \dot{\theta}_l^2) \\ &\quad + \frac{1}{2} (I_{wa} + I_{ra} \gamma^2) (\dot{\theta}_r^2 + \dot{\theta}_l^2) + (I_{wd} + I_{rd}) \dot{\phi}^2 \end{aligned} \quad (6)$$

$$U_g = M_g g l_g \cos \psi \quad (7)$$

$$D = \frac{1}{2} c_r (\dot{\theta}_r - \dot{\psi})^2 + \frac{1}{2} c_l (\dot{\theta}_l - \dot{\psi})^2 \quad (8)$$

Consequently, Lagrangian of the robot is $L = T_g + T_w - U_g$. The Lagrangian equation is expressed as

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} + \frac{\partial D}{\partial \dot{q}_i} = Q_i \quad i = 1, 2, \dots, n \quad (9)$$

where Q , q , and n represent a generalized force, coordinate, and coordinate number, respectively. Here, we define a generalized coordinate as $(x_c, y_c, \phi, \psi, \theta_r, \theta_l)$. Inserting L and D in Eq. (9) and solving it, the motion equation of the robot can be expressed as

$$M(q)\ddot{q} + H(q, \dot{q}) + V\dot{q} + G = E\tau \quad (10)$$

where $M(q) \in R^{6 \times 6}$ is an inertia matrix, $H(q, \dot{q}) \in R^{6 \times 1}$ is the coriolis and centrifugal force, $V \in R^{6 \times 1}$ is a viscous coefficient matrix, $G \in R^{6 \times 1}$ is the gravity force, and $E\tau \in R^{6 \times 1}$ is a control torque. Equation (10) represents the equation of motion in case all the constraints of the robot are holonomic in which integration is possible. However, a two-wheeled mobile robot as an I-PENTAR has non-holonomic constraints, which means it cannot move toward the direction perpendicular to a forward direction, as well as holonomic constraints. Therefore, it is necessary to modify Eq. (10) to the motion equation considering the non-holonomic constraints.

The motion equation with constraints is expressed as

$$M(q)\ddot{q} + H(q, \dot{q}) + V\dot{q} + G = E\tau + A(q)^T \lambda \quad (11)$$

where $A(q)$ is a matrix that represents the constraints of a robot and λ is a Lagrange multiplier. In order to derive $A(q)$, let us consider the constraints of a two-wheeled mobile robot.

As shown in Fig. 4, generally, there are three constraints in the robot, assuming that the wheels do not slip, which is given as

$$\dot{x}_c \sin \phi - \dot{y}_c \cos \phi = 0, \quad (12)$$

$$\dot{x}_c \cos \phi + \dot{y}_c \sin \phi = \frac{r_w}{2}(\dot{\theta}_r + \dot{\theta}_l) \quad (13)$$

$$\dot{\phi} = \frac{r_w}{2b}(\dot{\theta}_r - \dot{\theta}_l) \quad (14)$$

In the above equations, Eqs. (12) and (13) are non-holonomic constraints and Eq. (14) is a holonomic one. Subsequently, expressing them in a matrix, it is rewritten as

$$A(q)\dot{q} = 0 \quad (15)$$

where

$$A(q) = \begin{bmatrix} \sin \phi & -\cos \phi & 0 & 0 & 0 & 0 \\ \cos \phi & \sin \phi & b & 0 & -r_w & 0 \\ \cos \phi & \sin \phi & -b & 0 & 0 & -r_w \end{bmatrix}$$

Here, we define a matrix $S(q)$, which is composed of linear independent vector in the null-space of $A(q)$ as given below.

$$S(q) = \begin{bmatrix} \cos \phi & 0 & 0 \\ \sin \phi & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/r_w & b/r_w & 0 \\ 1/r_w & -b/r_w & 0 \end{bmatrix} \quad (16)$$

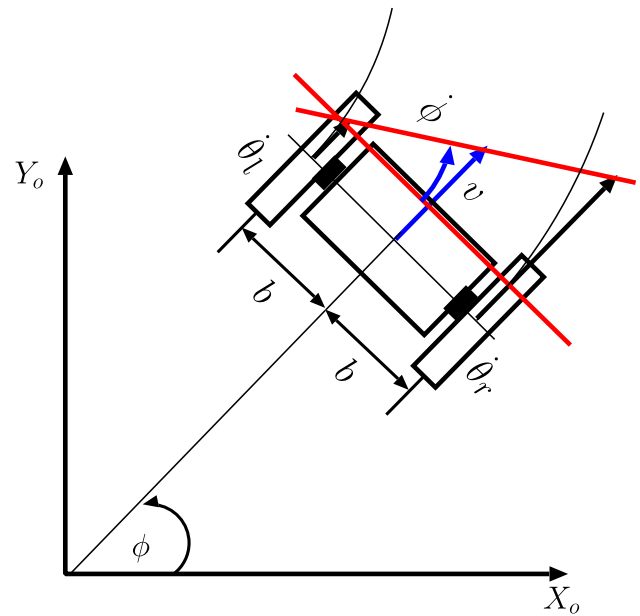


Fig. 4 Translational and rotational velocities of a wheeled mobile robot

In this case, $A(q)S(q) = 0$ since $S(q)$ is included in the null-space of $A(q)$. Moreover, since $\dot{q}$ always exists in the null-space of $A(q)$, defining a new velocity vector $v = [v \ \dot{\phi} \ \dot{\psi}]$, the following equation is satisfied.

$$\dot{q} = S(q)v \quad (17)$$

$$\ddot{q} = \dot{S}(q)v + S(q)\dot{v} \quad (18)$$

$$\hat{M}\dot{v} + \hat{H}(v, \dot{v}) + \hat{V}v + \hat{G} = \hat{E}\tau \quad (19)$$

where $\hat{M} = S(q)^T M(q) S(q)$, $\hat{H} = S(q)^T [M(q)\dot{S}(q)v + H(q, \dot{q})]$, $\hat{V} = S(q)^T V S(q)$, $\hat{G} = S(q)^T G$, and $\hat{E} = S(q)^T E$. The above equations represent the motion equations of the robot considering the holonomic and non-holonomic constraints.

3.3 Inverted mobile control

For the mobile control of I-PENTAR, in this paper, we adopt a state feedback control method and decide the feedback gain in the controller by an LQR method. In order to use the control method, first, Eq. (19) should be linearized since it is nonlinear now. Apart from the generalized coordinate used in the motion equation, a new generalized coordinate $q_v = [x \ \phi \ \psi]$ is introduced as an actual control variable. Then, the state variables in the system can be represented as $x = [q_v \ v]$. To linearize the system, it is approximated to around $x \cong 0$, and is expressed as $\sin(\cdot) = (\cdot)$, $\cos(\cdot) = 1$, $\sin^2(\cdot) = 0$, $(\dot{\cdot})^2 = 0$, and $(\dot{\cdot})(\dot{\cdot}) = 0$. On linearizing Eq. (19) using the expression, $\hat{H}$ disappears and the equation can be simplified as

$$\hat{M}\dot{v} + \hat{V}v + \hat{G} = \hat{E}\tau \quad (20)$$

where $\hat{M}$ and $\hat{G}$ are the linearized values. Therefore, the linearized state equation of the system is

$$\begin{aligned}\dot{\mathbf{q}}_v &= \mathbf{v} \\ \dot{\mathbf{v}} &= -\hat{M}^{-1}(\hat{V}\mathbf{v} + \hat{G}) + \hat{M}^{-1}\hat{E}\boldsymbol{\tau} \\ \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}\end{aligned}\quad (21)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & a_{43} & a_{44} & a_{45} & a_{46} \\ 0 & 0 & 0 & a_{54} & a_{55} & a_{56} \\ 0 & 0 & a_{63} & a_{64} & a_{65} & a_{66} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ b_1 & b_2 \\ b_3 & b_4 \\ b_5 & b_6 \end{bmatrix}$$

$$\text{where } a_{43} = \frac{M_g^2 l_g^2 r_w^2 G}{V_1}, \quad a_{44} = \frac{w_1 c_1}{V_1}, \quad a_{45} = \frac{b w_1 c_2}{V_1},$$

$$a_{46} = -\frac{r_w w_1 c_1}{V_1}, \quad a_{54} = -\frac{b c_2}{V_2}, \quad a_{55} = -\frac{b^2 c_1}{V_2},$$

$$a_{56} = \frac{b r_w c_2}{V_2},$$

$$a_{63} = -\frac{(M_g r_w^2 + 2m_w r_w^2 + 2(I_{wa} + I_{ra} \gamma^2)) M_g l_g G}{V_1},$$

$$a_{64} = -\frac{w_2 c_1}{r_w V_1}, \quad a_{65} = -\frac{b w_2 c_2}{r_w V_1}, \quad a_{66} = \frac{w_2 c_1}{V_1},$$

$$b_1 = b_2 = -\frac{r_w w_1}{V_1}, \quad b_3 = -b_4 = \frac{r_w b}{V_2}, \quad b_5 = b_6 = \frac{w_2}{V_1},$$

$$V_1 = M_g^2 l_g^2 r_w^2 - (M_g l_g^2 + I_{yy}) M_g r_w^2 - 2(M_g l_g^2 + I_{yy}) m_w r_w^2 - 2(M_g l_g^2 + I_{yy})(I_{wa} + I_{ra} \gamma^2),$$

$$V_2 = I_{zz} r_w^2 + 2(I_{wd} + I_{rd}) r_w^2 + 2m_w r_w^2 b^2 + 2(I_{wa} + I_{ra} \gamma^2) b^2,$$

$$w_1 = M_g l_g^2 + I_{yy} + M_g l_g r_w,$$

$$w_2 = M_g l_g r_w + M_g r_w^2 + 2m_w r_w^2 + 2(I_{wa} + I_{ra} \gamma^2),$$

$$c_1 = c_r + c_l, \quad c_2 = c_r - c_l.$$

The evaluation function of LQR is generally given as

$$J = \frac{1}{2} \int_0^a (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt \quad (22)$$

and the control input that minimizes the function is

$$\mathbf{u} = -\mathbf{K} \mathbf{x} \quad (23)$$

Deciding the optimal gain $\mathbf{K}$ by MATLAB using the parameter values in Table 3,

$$\mathbf{K} = \begin{bmatrix} -22 & 22 & -114 & -25 & 3 & -20 \\ -22 & -22 & -114 & -25 & -3 & -20 \end{bmatrix} \quad (24)$$

where $\mathbf{Q}$ is $\text{diag}([100 \ 100 \ 1,000 \ 0.1 \ 0.1 \ 0.1])$ and $\mathbf{R}$ is $\text{diag}([0.1 \ 0.1])$, and in such a case, the pole are $-11.6173 \pm 9.3825i$, $-1.1427 \pm 0.9721i$, $-7.4778 \pm 7.0801i$.

Table 3 Parameters of I-PENTAR

M_g (kg)	25.26
m_w (kg)	1.25×2
l_g (m)	0.4005
b (m)	0.16
r_w (m)	0.1
I_g (kgm ²)	(1.408, 1.39, 0.1082)
I_{wa} (kgm ²)	0.01475
I_{wd} (kgm ²)	0.0073
I_{ra} (kgm ²)	0
I_{rd} (kgm ²)	0
γ	1
c_r, c_l (Nm/rad/s)	(0.1, 0.1)
G (m/s ²)	9.81

4 Mobile experiment

In order to evaluate the control method described in the previous chapter, balancing, linear running, and steering experiments, which are considered as fundamental mobile motions of a wheeled inverted pendulum mobile robot, are conducted with I-PENTAR. In the experiments, I-PENTAR takes an upright-standing posture stretching its waist and extending its arms toward the ground. The position of CoG of the whole body, which is used for defining the robot's inclination angle, is calculated by a geometrical method based on the parameter values of each part.

4.1 Balancing control

In the balancing experiment, I-PENTAR keeps its balance initially and is pushed by an arbitrary force from the back. Figure 5 shows the angular velocity of the right wheel (up) and the inclination angle of CoG (down) in the experiment. We can see that the body of I-PENTAR is inclined backward while the wheel moves forward because of the force, and both of them return to original state after the force is removed. Then, in order to maintain balance, the wheel moves ahead when the body inclines forward, and if the body inclines backward, it moves backward. In the figure, it is observed that the wheel is vibrated forward and backward at 11 °/s to align the vibration motion of the inclination angle with 0.2 °/s. It is thought that this comes from the estimation error of the inclination angle of the body and the nonlinear factors in wheels, such as viscosity and coulomb friction.

4.2 Linear running control

Figure 6 shows the velocity (up) and displacement (down) of the robot in the linear running experiment. In the experiment, a troplized type velocity trajectory as shown in the

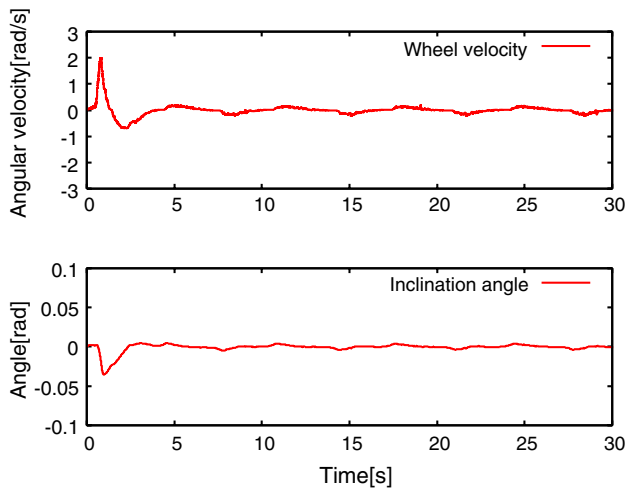


Fig. 5 Experimental result of a balancing motion, wheel velocity (*up*) and inclination angle (*down*)

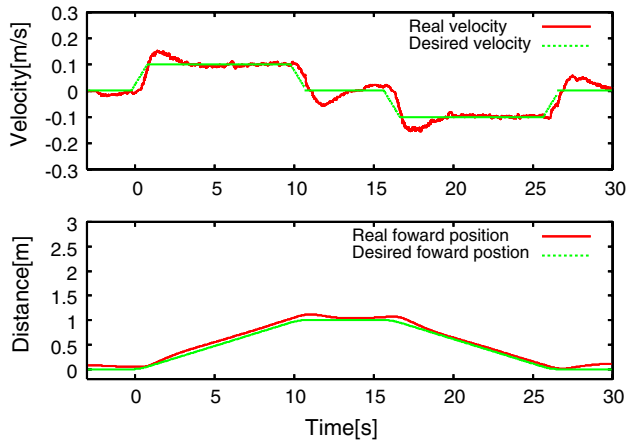


Fig. 6 Experimental result of a forward moving motion, forward velocity (*up*) and position (*down*)

upper figure is given, and the goal acceleration, velocity, and position are 0.2 m/s^2 , 0.1 m/s , and 1 m , respectively. From the figure, it is confirmed that the velocity of the robot in the uniform velocity zone follows the desired velocity. However, the tracking performance drops at the start of the acceleration and deceleration zone. The reason for this is that the gain related to linear running is very small to compensate the dynamic effect of the robot.

4.3 Steering control

In the steering experiment, the robot is operated such that it is rotated in a CCW direction on a spot. Figure 7 shows the profile of the wheel velocity (*up*) and the steering angle (*down*) in the experiment. Each wheel is rotated mutually and the rotation angle of the robot follows its desired angle. The tracking performance in a steering motion is high because the dynamic effect of the body is little during a steering motion.

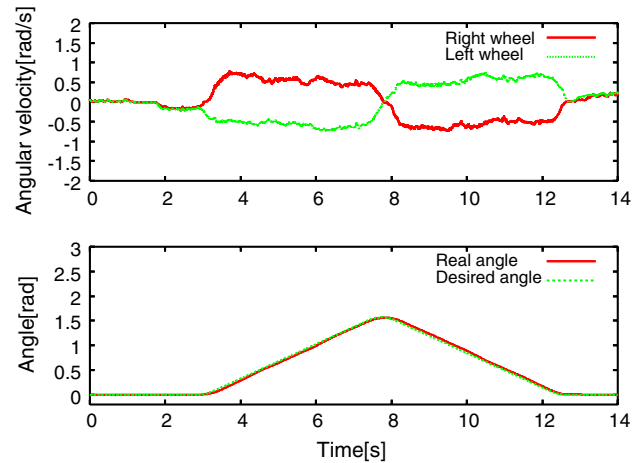


Fig. 7 Experimental result of a turning motion, wheel velocity (*up*) and turning angle (*down*)

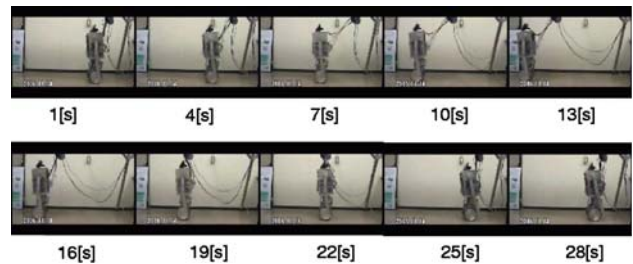


Fig. 8 Video clips of linear running motion

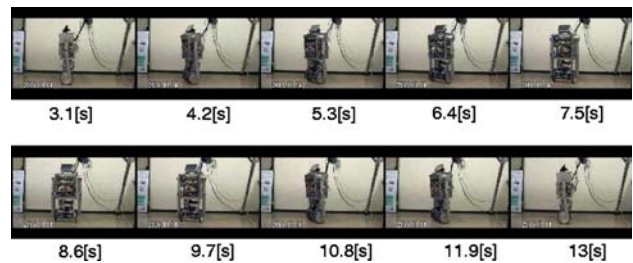


Fig. 9 Video clips of steering motion

5 Conclusion

In this paper, a human-assistant robot, I-PENTAR, which is aimed at achieving high safety and work capability simultaneously was proposed. I-PENTAR consists of a body with a waist, two arms designed for safety, and a two-wheeled inverted pendulum type mobile platform. It is able to attain a high work capability with its lightweight and low-power arms by using its posture with high-power efficiency and the instability that makes it possible to use self-weight. In this paper, in order to realize fundamental mobile motions, the state feedback control method based on the motion equation in consideration of non-holonomic constraints was applied. Through several experiments, it was confirmed that I-PENTAR could

run and steer stably, maintaining its balance by the controller. As a future work, we will try to consider dynamic effect of the robot in a mobile control in order to achieve agile motion.

References

1. Yamafuji K, Kawamura T (1988) Postural control of a monoaxial bicycle. *JRSJ* 7(4):74–79
2. Yamafuji K, Miyakawa Y, Kawamura T (1989) Synchronous steering control of a parallel bicycle. *JSME* 55(513):1229–1234
3. Yasui T, Yamafuji K (1991) Motion control of the parallel bicycle-type mobile robot which is composed of a triple inverted pendulum, 1st report, stability control of standing upright, ascending and descending of stairs. *JSME* 57(C)(538):114–119
4. Matsumoto O, Kajita S, Tani K (1990) Estimation and control of attitude of a dynamic mobile robot using internal sensors. *JRSJ* 8(5):37–46
5. Matsumoto O, Kajita S, Tani K (1991) Attitude estimation of the wheeled inverted pendulum using adaptive observer. In: 19th annual conference of RSJ, pp 909–910
6. Matsumoto O, Kajita S, Tani K (1998) Cooperative behavior of a mechanically unstable mobile robot for object transportation. *JSME* 64(C)(628):164–171
7. Ha Y, Yuta S (1994) Trajectory tracking control for navigation of self-contained mobile inverse pendulum. In: Proceedings of the 1994 IEEE/RSJ international conference on intelligent robotics and systems, pp 1875–1882
8. Ha Y, Yuta S (1995) Indoor navigation of an inverse pendulum type autonomous mobile robot with adaptive stabilization control system. In: 4th international symposium on experimental robotics (ISER)'95, pp 331–336, 1995–06.
9. Grasser F, D'Arrigo A, Colombi S (2002) JOE: a mobile, inverted pendulum. *IEEE Trans Ind Electron* 49(1):107–114
10. Pathak K, Franch J, Agrawal SK (2004) Velocity control of a wheeled inverted pendulum by partial feedback linearization. In: IEEE conference on decision and control, pp 3962–3967
11. Pathak K, Franch J, Agrawal SK (2005) Velocity and position control of a wheeled inverted pendulum by partial feedback linearization. *IEEE Trans Robot* 21(3):505–512
12. Kim YH, Lee D-Y, Kim SH, Kwak YK (2003) Emotional action simulation of 2 wheeled inverted pendulum mobile robot. The IASTED conference on modeling, simulation, and optimization, pp 266–271
13. Kim Y, Kim SH, Kwak YK (2005) Improving driving ability for a two-wheeled inverted-pendulum-type autonomous vehicle. In: Proceedings of the institution of mechanical engineers, part D. *J Automobile Eng* 220(2):165–175
14. Baloh M, Parent M (2003) Modeling and model verification of an intelligent self-balancing two-wheeled vehicle for an autonomous urban transportation system. In: The conference on computational intelligence, robotics, and autonomous systems, pp 1–7
15. Thibodeau BJ, Deegan P, Gruben R (2006) Static analysis of contact forces with a mobile manipulator. In: Proceedings of ICRA, pp 4007–4012
16. EMIEW, Hitachi, Ltd. <http://www.hqrd.hitachi.co.jp/merl/emiew.cfm>
17. Robin, Robo3, Ltd. <http://www.robo3.com/maink.htm>
18. Robovie 3, ART, Ltd. <http://www.irc.atr.jp/product/Robovie/robovie-2.html>